

Instructions: There is totally 1 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (10 points) Given the following differential equation:

$$\frac{dP}{dt} = 12P - 2P^2$$

(a) Find all critical points of the above autonomous differential equation and then determine whether it is stable or unstable for each of them. (Hint: Use derivative information)

$$f(P) = 12P - 2P^2 = 2P(6 - P)$$

$$f(P) = 0 \Rightarrow 2P(6 - P) = 0 \Rightarrow P = 0, P = 6 \leftarrow \text{two critical points}$$

$P'(t) < 0$ $P'(t) > 0$ $P'(t) < 0$ $P'(t) = f(P)$
 $\swarrow$ $\downarrow$ $\searrow$ $= 2P(6 - P)$
 $P < 0$ $0 < P < 6$ $P > 6$ $P = 0 \rightarrow \text{unstable}$
 $P = 6 \rightarrow \text{stable}$

(b) Solve for $P(t)$ the above differential equation with $P(0) = 2$ and find $\lim_{t \rightarrow +\infty} P(t)$.

$$\frac{dP}{dt} = 2P(6 - P)$$

$$\Rightarrow \int \frac{1}{P(6 - P)} dP = \int 2 dt$$

$$\frac{1}{6} \int \left(\frac{1}{P} + \frac{1}{6 - P} \right) dP = \int 2 dt$$

$$\ln|P| - \ln|6 - P| = 12t + C$$

$$\left| \frac{P}{6 - P} \right| = e^{12t + C}$$

$$\frac{P}{6 - P} = B e^{12t}$$

$$P(0) = 2 \Rightarrow \frac{2}{6 - 2} = B \Rightarrow B = \frac{1}{2}$$

$$\Rightarrow \frac{P}{6 - P} = \frac{1}{2} e^{12t}$$

$$2P = (6 - P) e^{12t}$$

$$(2 + e^{12t}) P = 6 e^{12t}$$

$$P(t) = \frac{6 e^{12t}}{2 + e^{12t}}$$

$$= \frac{6}{1 + 2 e^{-12t}}$$

$$\lim_{t \rightarrow +\infty} P(t) = \lim_{t \rightarrow +\infty} \frac{6}{1 + 2 e^{-12t}}$$

$$= \frac{6}{1 + 2 \cdot 0}$$

$$= 6$$